

MIAE215 Midterm Exam 2024 – Programming for Engineers Exam Guide PremiumCourse: [Mechanical, Industrial & Aerospace engineering \(MIAE 215\)](#) 378 documentsUniversity: [Concordia University](#)

2023/2024

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Concordia University
Department of Mechanical, Industrial and Aerospace
Engineering
MIAE215YY – Midterm Exam
Programming for Mechanical and Industrial Engineers

2024/02/22

Name: _____

Student Number: _____

1. What do the following code fragments print? Write your answer in the box.
(Marking: 1-a10%, 1-b10%, 1-c10%)

```
cout << (5%2) * int(1.5+2);
```

```
const double base = 2.75;  
cout << pow(base, 2);
```

```
for (int i = 1; i <= 10; i+=1) {  
    if (i%4 != 2) {  
        continue; }  
    else if (sqrt(i) == 3) {  
        break; }  
    cout << i << " "; }  
}
```

2. What is the output of the following programs? Consider the user input a value of 4321 where needed in this question. (Marking: 2-a10%, 2-b10%, 2-c10%).

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```
int num;  
int a=0;  
cout<<"Please Enter any number : ";  
cin >>num;  
while(num>0) {  
    num = num / 10;  
    a++;  
}  
cout<<a;
```

(a)4321

(b)321

(c)4

(d)3

```
int i , j=0;  
for( i =5;i >0;i/=2) {  
    j++;  
    cout<<j;  
}  
cout<<j;
```

(a)01234

(b)12345

(c)1233

(d)1234

```
int main() {  
    int result = 0;  
    for(int i = 1; i <= 10; i++) {  
        if (i % 3 == 0) {  
            for(int j = i; j <= 5; j++) {  
                result += j;  
            }  
        }  
    }  
    cout<<"The final answer is: "<<result <<endl;  
    return 0;  
}
```

3. What is the output of the following fragments of the programs? Write the answer in the box. (Marking: 3-a10%, 3-b10%, 3-c20%)

```
int main() {  
    int found = 1, count = 4;  
    if (!found || count == 0)  
        cout<<"safe"<<endl;
```



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```
cout<<"count = "<<count<<endl;
}
```

```
int main() {
    int i = 2;
    while(i <=20) {
        if(i % 3 == 0) {
            i += 2;
            continue;
        }
        cout<<i <<" ";
        if(i == 20) break;
        i += 2;
    }
    return 0;
}
```

```
int main() {
    int i;
    double x, y, z, q;
    q=x=y=z = 4;
    for(i = 0; i < 4; i++) {
        x=x * 1.5;
        y *= 2;
        z /= 2;
        q += 4;
    }
    cout<<"\nx = "<<x;
    cout<<"\ny = "<<y;
    cout<<"\nz = "<<z;
    cout<<"\nq = "<<q;
}
```

Mock exam

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